



ENSAE Paris – Institut Polytechnique de Paris

Algorithmic Trading - Pr. Olivier Guéant

Option Hedging with Smooth Market Impact

An article by Robert Almgren and Tianhui Michael Li (2016)

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April 2026

1. Introduction

In this report, we study the paper *Option Hedging with Smooth Market Impact* by Robert Almgren and Tianhui Michael Li [1]. Market impact models allowing reduced transaction costs when delta-hedging a portfolio for large traders is key to reduce loss considering the impacts of executing large positions. Numerous past events developed in the literature [2], [3] showed that for such large positions, delta-hedging strategies incur market impact by both price moves and volatility increase. Traders not taking into account those impacts may bear a loss when executing their large positions, or simply when trying to hedge their portfolios.

Historically, market impacts models considering price movements when executing large positions were first introduced by Leland [4] in 1985. Nonetheless, as seen with previously mentioned examples such as the sawtooth pattern observed in US equity markets in July 2012 [2], traders not using market impact models for their hedging strategies may result in the position being reversed if not executed properly [5], leading to more execution costs and potentially losses. In that particular case, the most plausible explanation for that sawtooth pattern on Coca-Cola, Apple, IBM and McDonald's stock prices was that a large trader was using a time-discrete delta-hedging strategy without market impact and executed their strategy every half hour.

The authors start with a simple market impact model to develop the hedging problem when holding large positions. They introduce their model with both a temporary impact depending on the rate of execution, and a permanent impact linked to the trade size. In the following section, we will review some of literature on which the paper is based, and the main framework they used to then establish what the key contribution of Almgren and Li was in their paper.

2. Motivation and Key Idea of Almgren and Li (2016)

Effects of transaction costs on Black-Scholes hedging using discrete-time model have been first introduced by Leland [4] and further developed in the literature [6], [7] : the main idea was to consider from the previous time interval an impact on the current time interval market price.

For their market impact modelling, a first choice made by the authors was using a finite mishedge penalty, instead of considering super-replication : it means that instead of considering a perfect hedge of the position in all cases, which may be costly and not representative of the market price, mishedge errors can occur but are penalized by either a utility function or a loss function.

Previous transactions costs models used trading frictions as a cost proportional to trade size, resulting in the form of tracking bands [8]. Instead, they use a market impact model decomposed in two parts : the permanent price impact, directly linked to trade size or the number of shares executed, and the temporary price impact, which corresponds to liquidity costs, linked to trade speed or the rate of execution.

Prior works in the literature on market impact models closer to this structure considered either only temporary market impact assumptions [9] [10], or temporary price impact [11] [12]. A similar framework designed by Guéant and Pu [13] included both market impacts, but made different assumptions on the model, opting for a more general solution to the problem using numerical computation.

The authors decided in particular to use mean-variance optimization, with the Hamilton-Jacobi-Bellman (HJB) equation. They made the assumption that their intraday model's position was market to market in the morning, not taking into account delivery.

The central contribution of Almgren and Li was developing a model that uses both permanent and temporary market impact, while allowing exact solutions given some restrictions in their framework. This allowed to then

qualitatively analyze their results to obtain insights on the effects on the market and the hedging strategy given those market impacts.

3. Summary of Methodology and Results

3.1. Model setup

They discuss the case of hedging a European option, using an objective function that would possibly include overnight risk. A critical analysis of their model can be found part 4. Critical Review. Their market impact model is presented as follows :

$$P_t = P_0 + \nu(X_t - X_0) + \sigma W_t$$

With $P_t, X_t, \nu, \sigma, W_t$ being respectively the fundamental price of the stock, the number of shares held by an agent, the coefficient of permanent impact, the absolute volatility of the fair value, and a standard arithmetic Brownian motion with filtration $\mathcal{F}_t$.

Trade rate verifies $\theta_t = \frac{dX_t}{dt}$, and as in Almgren and Chriss work in 2000 [14] they consider a price premium linear in θ_t , giving the effective trade price, with coefficient of temporary impact $\eta > 0$:

$$\tilde{P}_t(\theta_t) = P_t + \eta\theta_t$$

3.2. Hedging problem and stability conditions

The authors consider a European option over a time period of a trading day. The option value is denoted with $g(t, P_t)$. The delta and gamma are identified with:

$$\Gamma(t, p) = g''(t, p) \quad \Delta(t, p) = -g'(t, p) \quad (\text{minus sign as trader is long the payoff})$$

Γ is assumed to be uniformly bounded above and below, and Lipschitz in p with constant independent of t . This means the authors only consider the option outside the expiry day, as around the strike at maturity Γ diverges. Since Γ is in that case negative, the authors add a constraint to make sure the option hedging doesn't change the market dynamics:

$$\exists G > 0 \quad s.t. \quad \forall(t, p), 1 + \nu\Gamma(t, p) \geq G$$

With a Black-Scholes hedging argument, without market impact, using the Feynman-Kac equation given the assumptions regarding interest rates then integration by parts, they obtain:

$$R_T = R_0 + \sigma \int_0^T Y_t dW_t + \nu \int_0^T Y_t \theta_t dt - \eta \int_0^T \theta_t^2 dt$$

This would represent the final wealth, marked-to-market at the end of the day, without market impact from converting the position to cash.

The mishedge error, or the net delta exposure, is : $Y_t = X_t - \Delta(t, P_t) = X_t + g'(t, P_t)$. A super-replication strategy would yield $Y_t = 0$. The terms are respectively the initial wealth, the stochastic risk from market, the permanent impact and the temporary impact.

3.2.1 Mean-variance criterion with overnight risk

The authors adopt a mean-variance criterion. Since all the terms in R_T are random, the authors decided that the variance of R_T can be approximated with "small-portfolio" assumption [15] with its the main source of uncertainty, the price motions (stochastic risk from market):

$$\mathbb{V}[R_T] \approx \sigma^2 \mathbb{E} \left[\int_0^T Y_t^2 dt \right].$$

Since the position is marked to market at tomorrow's open T^* rather than today's close T , overnight risk is included. The full control problem is:

$$\inf_{\theta \in \Theta} \mathbb{E} [\lambda (Y_T \Pi - \xi)^2 + \lambda \sigma^2 \int_0^T Y_t^2 dt - \nu \int_0^T Y_t \theta_t dt + \eta \int_0^T \theta_t^2 dt]$$

where $\lambda > 0$ is the variance penalty, $\Pi = P_{T^*} - P_T$ is the overnight price move (variance Σ^2), and $\xi = \int_T^{T^*} [\Delta(t, P_t) - \Delta(T, P_T)] dP_t$ captures the inability to rebalance during the night. This overnight term introduced by the authors penalizes any residual mishedge at the close of trading.

3.3. HJB equation and general optimal strategy

By dynamic programming, the value function $J(t, p, y)$ satisfies a Hamilton-Jacobi-Bellman (HJB) equation. The infimum over θ is a quadratic minimization. The optimal strategy is:

$$\theta_t = -\frac{1}{2\eta} [(1 + \nu\Gamma)J_y + \nu J_p - \nu y].$$

Where $J_p \equiv \partial_p J$. If $\nu = 0$, the trader adjusts the position at a speed driven by $J_y = \partial_y J$, dampened by η . If $\nu > 0$, the term νJ_p captures the effect of permanent impact on the future price, while $\nu\Gamma J_y$ accounts for the modification of the delta target induced by each trade. The last term $\frac{\nu y}{2\eta}$ represents an internal arbitrage opportunity: since the position is marked to market via the public price, the trader can increase the book value by trading in the direction of the position. This effect is bounded by risk aversion λ .

3.4. Case 1: constant gamma with permanent impact

The approximation $\Gamma \equiv \text{const}$ is reasonable considering that the time horizon is a single day supposed to be far from expiry, which eliminates the dependence on both t and P_t , as well as reducing the problem to a stochastic linear regulator. The condition on G then simplifies to the constant $G = 1 + \nu\Gamma > 0$.

3.4.1 Explicit solution and Riccati equation

The constant Γ makes the value function independent of p and quadratic in y :

$$J(t, y) = \frac{1}{2} A_2 (T - t) y^2 + A_0 (T - t)$$

The coefficient A_2 satisfies the scalar Riccati equation:

$$\frac{dA_2}{dt} = 2\lambda\sigma^2 - \frac{(GA_2 - \nu)^2}{2\eta} \quad A_2(0) = 2\lambda\Sigma^2, \quad G = 1 + \nu\Gamma$$

which admits two equilibrium values $A_2^\pm = \frac{\nu \pm 2\sqrt{\eta\lambda\sigma^2}}{G}$. Under the stability condition $2\lambda\Sigma^2 > A_2^-$, the coefficient converges to the stable point A_2^+ . The explicit solution is:

$$A_2(\tau) = \frac{\nu + 2\eta\kappa h(\kappa G\tau)}{G} \quad \text{where} \quad \kappa = \sqrt{\frac{\lambda\sigma^2}{\eta}}, \quad h(s) = \frac{1-u}{1+u}, \quad u = \frac{1-h_0}{1+h_0}e^{-2s}, \quad h_0 = \frac{G\lambda\Sigma^2 - 2\nu}{\eta\kappa}$$

The optimal trading intensity takes the form:

$$\theta_t = -\kappa h(\kappa(T-t)G)Y_t$$

The result obtained by the authors is such that the trader continuously targets the Black-Scholes delta at a speed proportional to the mishedge Y_t and the urgency parameter $\kappa = \sqrt{\frac{\lambda\sigma^2}{\eta}}$ as in Almgren and Chriss paper [14]: higher volatility or risk aversion increases κ parameter, whereas higher illiquidity decreases it. The dependence on Γ enters only weakly through $G = 1 + \nu\Gamma$, as the author assumed $|\nu\Gamma| \ll 1$.

Far from expiry ($\kappa(T-t) \gg 1$), $h \equiv 1$ and the rule simplifies to $\theta_t = -\kappa Y_t$. Near the close, h_0 governs the regime:

- $h_0 > 1$: overnight risk dominates, trading intensity increases near the close: the trader is willing to pay more in temporary impact to reduce overnight risk of the mishedged position, as explained by the authors
- $0 < h_0 < 1$: permanent impact partially dampens trading without reversing it
- $h_0 < 0$: permanent impact dominate, the trader reduces or even reverses hedging near the close, exploiting the mark-to-market arbitrage.

In practice, $h_0 > 1$ is the most realistic regime for a normal trading day.

3.4.2 Collective effects on realized volatility

When all hedgers collectively hold a net gamma Γ and trade optimally with $h \equiv 1$, the aggregate mishedge follows an Ornstein-Uhlenbeck process:

$$Y_t = \sigma\Gamma \int_0^t e^{-\tilde{\kappa}(t-s)} dW_s, \quad \tilde{\kappa} = G\kappa$$

and the public price process becomes:

$$P_t = P_0 + \frac{\sigma}{G} \left(W_t + \nu\Gamma \int_0^t e^{-\tilde{\kappa}(t-s)} dW_s \right)$$

The effective volatility measured over an interval δt is:

$$\sigma_{\text{eff}}^2(\delta t) = \left(\frac{\sigma}{G} \right)^2 \left(1 + (2 + \nu\Gamma)\nu\Gamma \frac{1 - e^{-\tilde{\kappa}\delta t}}{\tilde{\kappa}\delta t} \right)$$

with limits $\sigma_{\text{eff}} \rightarrow \sigma$ for $\tilde{\kappa}\delta t \ll 1$ and $\sigma_{\text{eff}} \rightarrow \frac{\sigma}{G}$ for $\tilde{\kappa}\delta t \gg 1$.

If $\Gamma > 0$ (net long), $G > 1$ and realized volatility is reduced: hedgers act as mean-reverting agents, buying when the price rises and selling when it falls (option pinning effect). If $\Gamma < 0$ (net short), $G < 1$ and volatility is amplified, explaining historical episodes such as the October 2014 Treasury flash crash [2].

The characteristic time scale $\tilde{\kappa}^{-1}$ is set by temporary impact η and risk aversion λ , while the amplitude of the modification depends solely on permanent impact ν and the net position Γ . This extends Lions and Lasry works [11] [12], who obtained a modified volatility at all time scales due to the absence of temporary impact. Here, the temporary impact added by the authors introduces a minimum time scale $\tilde{\kappa}^{-1}$ below which the market remains efficient, consistent with empirical evidence.

3.5. Case 2: general gamma without permanent impact

In the case of a general $\Gamma(t, p)$ function, the authors removed the permanent impact to obtain an analytical solution. The value function satisfies this times :

$$J(t, p, y) = \frac{1}{2}A_2(T-t)y^2 + A_1(T-t, p)y + A_0(T-t, p)$$

Where A_2 verifies the same Ricatti equation as with constant Γ , A_0 and A_1 verifying other PDEs. We will skip the resolution, to go directly to the result obtained by the authors, who get the optimal control :

$$\theta_t = -\kappa h(\kappa(T-t))(Y_t - \bar{Y}_t), \quad \bar{Y}_t = \frac{\lambda}{\eta\kappa \sinh(\kappa(T-t)) + \lambda\Sigma^2 \cosh(\kappa(T-t))} F(T-t, P_t)$$

With F verifying the heat equation $\frac{dF}{dt} = \frac{1}{2}\sigma^2 F''$ with initial condition $F(0, p) = \mathbb{E}[\Pi\xi | P_T = p]$.

Far from the close of trading T , we get the same trade intensity, as $\bar{Y}_t$ is close to 0. However, the authors explain the terminal asymmetry, where the offset from the perfect Black-Scholes hedge $Y_t = 0$ by $\bar{Y}_t$ being significant: if Γ is increasing in p near P_T , Δ is concave, and the authors obtain that $\bar{Y}_t < 0$, meaning the trader should end the trading day while being underhedged, as the overnight moves are more significant if P_t increases than if P_t decreases. The authors explain the situation is reversed if Γ is decreasing in p near P_T .

Without permanent impact, the previously targeted delta-hedge from Black-Scholes from the constant Γ case is no longer targeted, but instead having an offset whose sign depends on the convexity of Δ .

3.6. Unconditionally stable discretization

The naive rule $\theta_k = -\kappa h Y_k$ yields $Y_{k+1} = (1 - G h \kappa \delta) Y_k$, which is unstable whenever $\kappa \delta > 2$, which is exactly what produced the sawtooth pattern mentioned in the introduction [3]. For small temporary impact η , the relaxation rate κ may be large, making this constraint very restrictive.

The authors derive the fully optimal discrete-time rule by solving the control problem exactly over each interval $[t_k, t_{k+1}]$, accounting for the continuous evolution of Y_t under a constant rate θ_k :

$$\theta_k = -\frac{GA_{k+1} - \nu + \lambda\sigma^2 G\delta}{2\eta + (GA_{k+1} - \nu)G\delta + \frac{2}{3}\lambda\sigma^2 (G\delta)^2} Y_k$$

with $A_N = 2\lambda\Sigma^2$ and the backward recursion $A_k = F(A_{k+1})$, where:

$$F(A) = A + \left(2\lambda\sigma^2 - \frac{(GA - \nu + \lambda\sigma^2 G\delta)^2}{2\eta + (GA - \nu)G\delta + \frac{2}{3}\lambda\sigma^2(G\delta)^2} \right) \delta$$

The authors prove that under the same stability condition as the continuous case, the sequence (A_k) is well-defined, non-negative, and converges to a fixed point for any time step δ , with no restriction on $\kappa\delta$. In the limit $\delta \rightarrow 0$, the discrete rule recovers the continuous-time optimal strategy.

4. Critical Review - Conclusion

As mentioned previously in part 2 when detailing the framework and the main idea of Almgren and Li in their paper, introducing a smooth market and continuous market impact allowed a more realistic result compared to the tracking bands obtained with proportional transaction costs [8]. While their framework are more restrictive to allow exact analytical solutions, introducing both temporary and permanent market impacts, as well as overnight risk, was new compared to prior works [9] [10] [11] [12]. Furthermore, having a framework that allowed exact analytical solutions was a plus compared to Guéant and Pu's more general numerical solutions [13].

Given the single day horizon, their assumptions of not taking into account the dividend nor the interest rate is simplifying the computation, as is opting for an Arithmetic Brownian motion instead of a Geometric Brownian one. However, if the daily volatility is very high given, the model might yield negative stock prices when doing numerical simulations, which is unrealistic. The short time horizon allows this assumption to be valid. However, since Γ must be bounded, this model is not suitable for the study of the expiry date. The bound $\exists G > 0 \quad s.t. \forall(t, p), 1 + \nu\Gamma(t, p) \geq G$ implies that the the option scale cannot be too large, which in their case means that they only consider intermediate market impact. Since they wanted to model market impact as seen with sawtooth pattern [3], this framework doesn't study such large positions close to expiry, where Γ is not constant.

A first possible extension of their work would be to use a multi-asset framework, as seen in the course with a simpler case, based on Almgren-Chriss framework [14] as in Almgren and Li did in this paper [1] with a linear market impact on the price variation: $h^i(v_t^i) = \eta_i v_t^i$, without overnight risk, where some restrictive framework gave us an explicit solution for optimal execution given market impact:

$$z(t) = \sinh\left(\sqrt{\frac{\gamma\Delta}{2}}(T-t)\right) \sinh^{-1}\left(\sqrt{\frac{\gamma\Delta}{2}}T\right) z(0)$$

With γ and Δ being respectively the risk aversion coefficient matrix in the mean-variance optimization problem, and Δ being the eigen values matrix of the obtained Euler-Lagrange equation from a mean-variance optimization problem. The outline of the above solution for this mean-variance optimization problem is as following:

$$\begin{cases} dq_t = v_t dt \in \mathbb{R}^d, dS_t = V dW_t & W_t \text{ being a } \mathbb{R}^d - \text{brownian motion} \\ dX_t = \sum_{i=1}^d v_t^i (S_t^i + \eta_i v_t^i) dt \\ \implies X_T = X_0 + q_0^T S_0 - \int_0^T \sum_{i=1}^d \eta_i (v_t^i)^2 dt + \int_0^T (q_t')^T V dW_t \\ \mathbb{E}[X_T] = X_0 + q_0^T S_0 - \int_0^T \sum_{i=1}^d \eta_i (v_t^i)^2 dt \\ \mathbb{V}[X_T] = \int_0^T (q_t')^T \Sigma q_t' dt \end{cases}$$

Expressing this problem using mean-variance optimization leads to:

$$\max_q \mathbb{E}[X_T] - \frac{\gamma}{2} \mathbb{V}[X_T] \iff \min_q \int_0^T (\frac{\gamma}{2} q_t^T \Sigma q_t + (q_t')^T \Theta q_t') dt$$

For the resolution to the final equation leading to the above expression of $z(t)$:

$$\left\{ \begin{array}{l} \text{Euler Lagrange : } \gamma \Sigma q_t = 2 \Theta q_t'', q(0) = q_0, q(T) = 0 \\ \Sigma = V V^T, y_t = V^T q_t, \Theta = \text{Diag}((\eta_i)_{1 \leq i \leq d}) \text{ symmetric matrix} \\ \frac{\gamma}{2} V^T \Theta^{-1} V y_t = y_t'' \\ \text{Spectral theorem : } V^T \Theta^{-1} V = \Omega \Delta \Omega, \text{ with } \Delta = \text{Diag}((\delta_i)_{1 \leq i \leq d}) \\ \implies \frac{\gamma}{2} \Delta z = z'', z_t = \Omega^T y_t, \text{ thus the result.} \end{array} \right.$$

The multi-asset optimal execution strategy established extended from Almgren and Li's work in the single asset case [1] was developed in a paper published by Bergault, Drissi, and Guéant in 2022 [16], using a restrictive framework considering Ornstein-Uhlenbeck Dynamics, but obtaining exact analytical results.

As seen in the course, another extension could be such that the execution cost model introduces non-linearity for the transaction cost, with $h(v_t) = \psi + \eta \sqrt{|v_t|}$ instead of the linear form $h(v_t) = \eta v_t$. They mention previous work that suits more empirical data using non linear function [17]. Introducing participation ratio can also be taken into account since the authors consider a large trade size: $v_t \rightarrow \frac{v_t}{V_t}$ with V_t being the total market volume, v_t being the trade intensity at time t. However, it would exit the simpler framework they opted for to obtain analytical solution. This kind of extensions would be more suitable for the work of Guéant and Pu [13] which uses numerical resolution.

Extending the time horizon to multiple days would allow a broader view of the market impact, but as mentioned before, most of the assumptions made by the authors for small time horizons would not apply anymore.

Introducing a stochastic volatility instead of the constant volatility used would allow to capture the vega risk, mentioned in the introduction of the paper, but absent in their model. The stochastic volatility Heston model would allow to introduce the dynamics of both the underlying and the volatility using a 2-dimensional correlated brownian motion.

As the general Γ framework completely removed any permanent impact, the framework expanded by Guéant and Pu [13] keeps both impacts while having a non-constant Γ function. We see the usual tradeoff between model analytical simplicity and realism that Almgren and Li had to face when writing their paper.

References

- [1] Robert Almgren and Tianhui Michael Li, *Option Hedging with Smooth Market Impact*, Market Microstructure and Liquidity, 2(1), 2016.
- [2] Robert Almgren, *Treasury price swings on October 15, 2014.*, Technical report, Quantitative Brokers, 2015.

- [3] I. Hwang, W. Kisling, and N. Mehta, *Investors whipsawed by price swings in IBM, Coca-Cola.*, Bloomberg, 2012.
- [4] H. E. Leland, *Option pricing and replication with transactions costs*, J. Finance, 40(5):1283–1301, 1985.
- [5] C.-A. Lehalle and M. Lasnier, *What does the saw-tooth pattern on US markets on 19 July 2012 tell us about the price formation process*, Technical report, CA Chevreux Quantitative Research, Aug. 2012.
- [6] Y. M. Kabanov and M. M. Safarian, *On Leland’s strategy of option pricing with transactions costs*, Finance Stoch., 1(3):239–250, 1997.
- [7] Y. Zhao and W. T. Ziemba, *Hedging errors with Leland’s option model in the presence of transaction costs*, Finance Research Letters, 4:49–58, 2007
- [8] M. H. A. Davis and A. R. Norman, *Portfolio selection with transaction costs*, Math. Oper. Res., 15(4):676–713, Nov. 1990
- [9] Gârleanu and L. H. Pedersen, *Dynamic trading with predictable returns and transaction costs*, J. Finance, 68(6), Dec. 2013.
- [10] C. G. Rogers and S. Singh, *The cost of illiquidity and its effects on hedging*, Math. Finance, 20(4):597–615, Oct. 2010.
- [11] P.-L. Lions and J.-M. Lasry, *Towards a self-consistent theory of volatility*, J. Math. Pures Appl., 86:541–551, 2006.
- [12] P.-L. Lions and J.-M. Lasry, *Large investor trading impacts on volatility*, Ann. Inst. H. Poincaré Anal. Non Linéaire, 24(2):311–323, 2007
- [13] O. Guéant and J. Pu, *Option pricing and hedging with execution costs and market impact*, Math. Finance, 2015. doi: 10.1111/mafi.12102.
- [14] R. Almgren and N. Chriss, *Optimal execution of portfolio transactions*, J. Risk, 3(2):5–39, 2000.
- [15] R. Almgren and G. Burghardt, *A window into the world of futures market liquidity*, Research snapshot, Newedge Prime Brokerage, March 2011
- [16] P. Bergault, F. Drissi, O. Guéant, *Multi-asset Optimal Execution and Statistical Arbitrage Strategies under Ornstein-Uhlenbeck Dynamics*, SIAM J. FINANCIAL MATH, 2022. Vol. 13, No. 1, pp. 353–390
- [17] R. Almgren, C. Thum, E. Hauptmann, and H. Li, *Equity market impact*, J. Risk, 18(7,July):57–62, 2005